

infOrigami: A Computer-aided Design Method for Introducing Traditional Perforated Boneless Lantern Craft to Everyday Interfaces

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ABSTRACT

The traditional perforated boneless lantern craft is a three-dimensional (3D) lamp fabrication method in which paper sheets are folded without any skeleton, and often accompanied by hole-like patterns. It has been extensively applied throughout history using various means and is favored because of its low cost and environmental friendliness. However, the traditional craft requires a complicated manual production process and accumulated craft expertise, which limits its aesthetics and daily use-value. We propose a computer-aided design method to customize a 3D lamp with decorative patterns from paper pieces. The key idea is to establish an automatic workflow for producing 3D objects with visual information that is similar to the lantern tradition but without overloaded manual efforts. Our user tests and results demonstrate the potential for DIY everyday interfaces for creating personalized aesthetics, enhancing the visualization of information, and promoting multi-person handicraft activity.

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CCS CONCEPTS

• **Human-centered computing**; • **Human computer interaction (HCI)**;

KEYWORDS

Hybrid Craft, CAD, Traditional Boneless Lantern, Everyday Interfaces

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1 INTRODUCTION

As an extension of human manual practice, the hybrid craft is becoming increasingly favored by human-computer interaction (HCI) researchers rather than completely replacing manual tasks by automation. Computer-aided craft design is a common method used to reduce the difficulty for novices while improving the design space for shapes, patterns, and other visual expressions. For example, Flower Jelly Printer [4] allowed users to design and create their own flower jellies. KnitUI [3] presented an accessible machine-knitted user interface for customizing colors, sizes, positions, and shapes. 3D Pen + 3D Printer [7] proposed techniques and a system that empowers users to explore creations while doodling. Papercut [2] proposed a design tool to help users to craft paper-cut folk art. In this paper, we investigated the Chinese traditional perforated boneless lantern craft which is a 3D lamp fabrication method in which

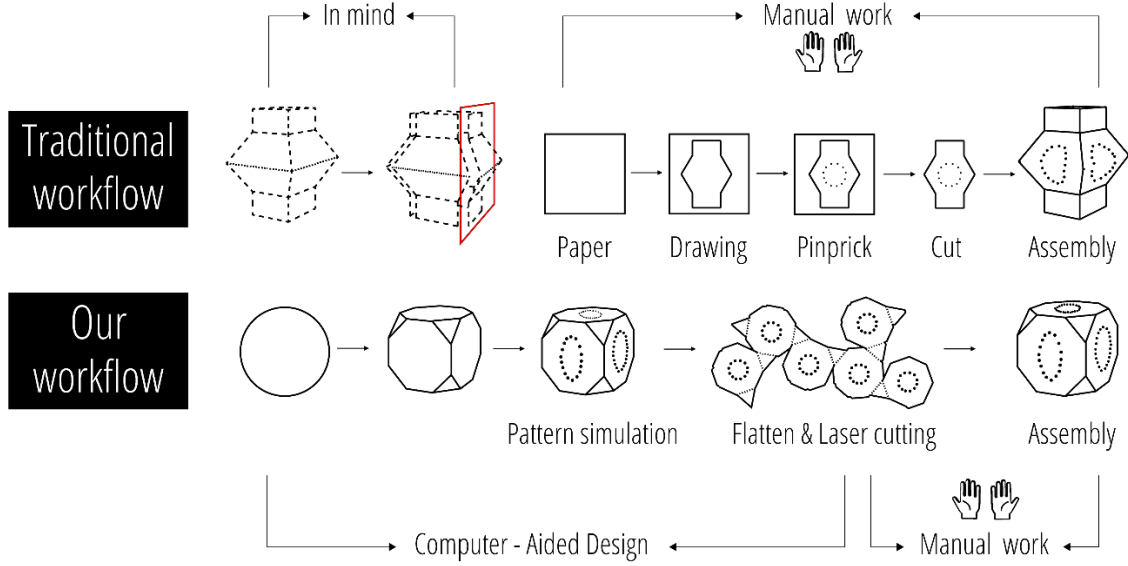


Figure 1: Workflow comparison.

paper sheets are folded without any skeleton, often accompanied by hole-like patterns. It has been extensively applied throughout history using various means and is favored because of its low-cost and environmental friendliness. However, the traditional craft requires a complicated manual production process, and accumulated craft expertise, which limits its aesthetics and daily use value [9]. We found that traditional perforated boneless lanterns combine a shape and pattern into a functional artifact, which has a new opportunity to be further developed as an everyday physical interface.

Simultaneously, as one of the fastest and most common types of digital fabrication technology, laser cutting is typically used by HCI researchers to make 3D prototypes. For example, LaserOrigami [5] provided a method to fabricate 3D shapes during laser cutting, Fabricaide [6] enabled users to proactively tailor their designs to machine-cut objects. LamiFold [1] allowed users to laser cut our functional mechanisms. WeaveMesh [8] enabled users to produce 3D objects in a mesh structure with flat pieces. We use laser cutting as our accessible fabrication method to reduce non-creative tasks, thereby allowing users to pay more attention to creative design, the manual assembly process, and interaction with an everyday lamp.

We propose a computer-aided design tool called infOrigami that can help novice users to customize 3D lamps using DIY decorative patterns from paper pieces. Our key idea is to establish an automatic workflow for producing 3D objects with visual information that is similar to the lantern tradition, but without the overloaded manual efforts. Our work contributes an accessible fabrication method for crafting everyday interfaces, which provides a potential approach to creating personalized aesthetics, enhancing the visualization of information, and promoting multi-person handicraft activity.

2 TRADITIONAL PERFORATED BONELESS LANTERN

The traditional workflow can be roughly divided into five steps, which are always required two to four days to complete an entire lantern. First, bind the draft paper and the lantern paper to prevent shifting. Second, outline the model on the white draft paper. Third, create pinprick patterns and trace an outline on the lantern paper. Fourth, make an incision on the paper along the outline. Finally, assemble the pieces of the lantern. Difficulties are typically encountered in the traditional workflow during the needling process, which requires a high degree of skill to achieve the equivalent strength each time. Additionally, verifying the feasibility of the innovative design of the lantern body shape requires a great deal of time and effort (Figure 1).

In this paper, we propose a computer-aided design method to facilitate the fabrication of customized 3D lamps with decorative patterns from paper pieces. Our method reduces non-creative tasks by replacing the manual work with parametric design software and laser cutting.

3 COMPUTER-AIDED DESIGN TOOL: INFORIGAMI

We developed our infOrigami design tool on top of the Rhino and Grasshopper plugins, which allows an inverse design for a lamp.

3.1 Inverse Design Workflow

As Figure 2 shows, users can a) import a mesh model and b) adjust the number of mesh surfaces to maintain the balance the model complexity and model shape, then c) design the desired pattern by adjusting the parameter sliders, d) flatten the model to gain the paper sheets for laser cutting, and finally f) assemble the paper sheets into a whole lamp.

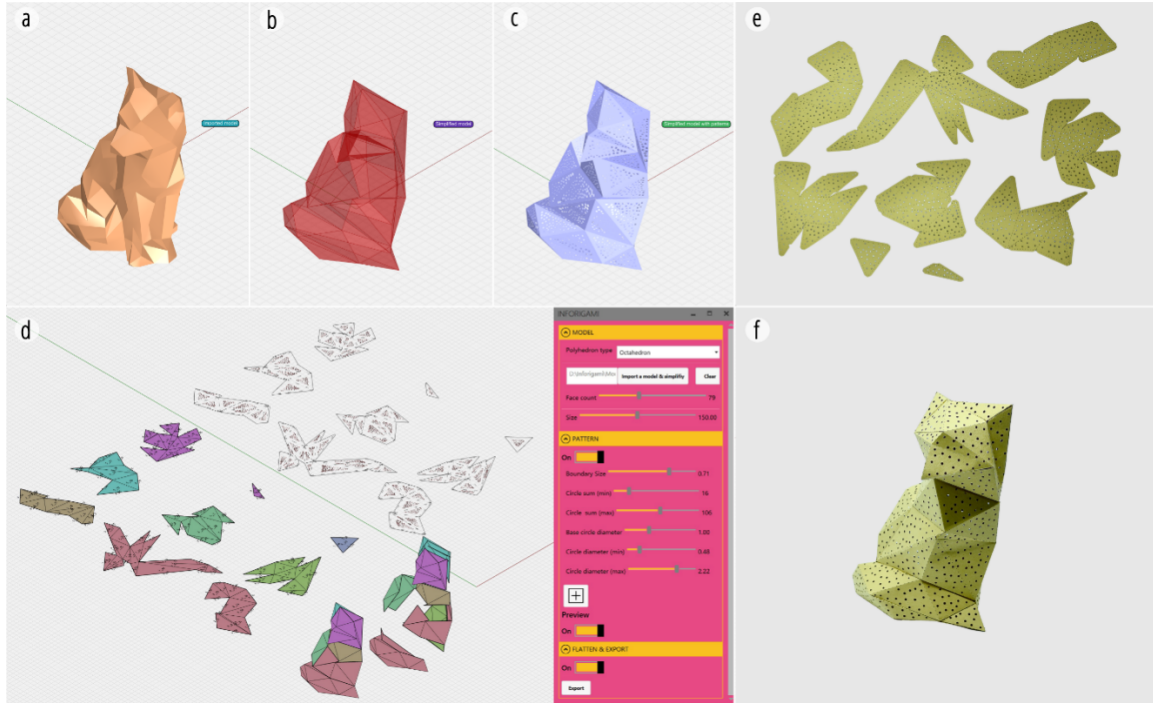


Figure 2: Inverse design workflow using infOrigami.

3.2 User Interfaces

3.2.1 Simplify 3D model. Users can generate custom pattern paper sheets using two approaches. Our software provides a library that contains a branch of polyhedrons. Users can select one of them as their target base model, or import another model by clicking the "import" button. The face-count slider is designed to adjust the number of faces generated on the mesh. The process of unfolding then converts the 3D mesh into 2D that is ready for laser cutting and crafting. After setting the desired assembling complexity and target resolution, users can use this feature to customize the 2D mesh accordingly. Normally, a face count such as 100 would generate a satisfactory result.

3.2.2 Generate Pattern. The perforation of the surface origami lantern determines the illumination effect. Our system computes and sorts all faces, and generates the maximum circles on the largest face and minimum circles on the smallest face. Different numbers of circles are generated according to their surface area. In addition to the circle sum, the user can also adjust the parameter of the circle diameter. Users can set a number as the base circle diameter, and the closer the circle is to the center of its surface, the larger its diameter and vice versa. The circle ranges are determined by the last two parameters. After users set the proper settings, they can pull the toggle to preview the model with the patterns. However, this procedure requires some time, maybe 10 to 15 seconds; hence, users need to be patient if the computer is freezing. Finally, users can import another type of pattern by clicking the add button.

3.2.3 Flatten & export. Users pull the toggle to flatten the model into several pieces of origami sheet and click the export button to

generate models for laser cutting. Generally, the number of pieces is determined by the face count of the model. Additionally, the fabrication process of the lantern involves rigorous matching, which is facilitated by the software output. Our system provides the color match, number specification, and splitting models to facilitate users to assemble the origami. Different colors are assigned for different parts of the model, which are the same colors as the sheet pieces. As a result, users can find a specific piece of the model by matching the color. Meanwhile, the sheet pieces are marked with numerical labels to help users to find the matching edges for gluing. Specifically, the center of each face of the flattened sheet piece contains a numerical label, which consists of two numbers: the first number represents the sheet piece index and the second number represents the face index. Additionally, some sides of a face may also contain labels, which represent the piece index and face index of the face to which they are attached. Thus, users can use this method for gluing.

4 USER STUDY

We invited three users with different backgrounds to design and fabricate their own lamps using infOrigami system. Before the formal trial, we trained users on this system for 30 minutes. Then we invited them to use infOrigami system to design their lamps. After the fabrication process, we asked them to provide feedback on the system.

4.1 Fabrication Test

To better evaluate the approach in practice, we recorded the time consumed by laser cutting and manual assembly. The results demonstrated that the laser cutting time was proportional to the number

Table 1: Participant information

Participant	P1 (Father & Daughter)	P2	P3
Age	34 & 3	21	28
Experience	Novice without pinprick experience.	Novice without pinprick experience.	has experienced the workflow of traditional perforated boneless lantern craft.
Software ability	Skillful	Skillful	Basic ability

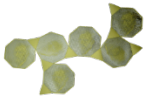
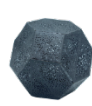
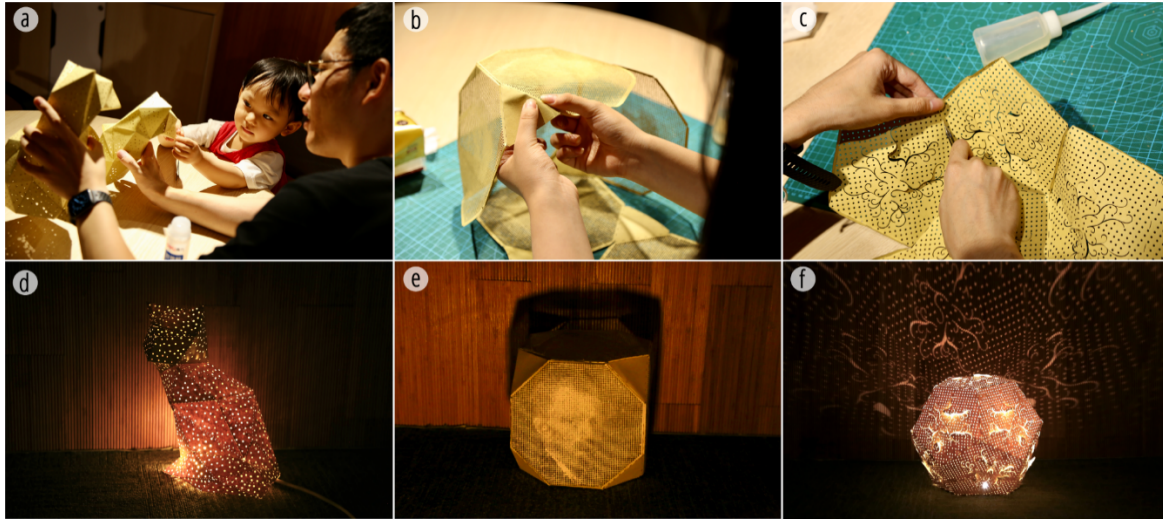
Laser Cutting	Time of Laser Cutting	Assembly	Time of Assembly	Laser Cutting	Time of Laser Cutting	Assembly	Time of Assembly
	4 h		30 min		48 min		20 min
	20 min		60 min		40 min		40 min

Figure 3: The time of laser cutting and assembly.**Figure 4: (a-c) The manual processes of (d-f) the final lamps.**

of patterns, and the assembly time was proportional to the number of edges. All the cases shown in Figure 3 were achieved in 5 hours.

4.2 Everyday Interfaces

4.2.1 Creating personalized aesthetic lamps. Compared with the fixed shapes and patterns of traditional craft lanterns, infOrigami provides users with the opportunity to design a customized lantern with attractive models and patterns that are in line with the user's aesthetic. "The software is a convenient tool for generating lamp models and patterns." (P2)

4.2.2 Enhancing the visualization of information. Light makes objects visible and is also a medium for transmitting information. Traditional boneless lanterns transmit only a small amount of information in the default limited patterns. By contrast, infOrigami can create different personal information carriers. Therefore, our method enhances the visualization of information heavily. "Personalized lamps can represent more individual wishes." (P2)

4.2.3 Promoting multi-person handicrafts activity. For some complex models, it takes rather a long time to assemble multiple pieces into a whole part, whereas a multi-person activity could shorten the

time notably. Furthermore, accompanying others and cooperating during the crafting process is a wonderful experience for all participants, obviously. "It can enhance the parent-child relationship by making this together." (P1)

5 CONCLUSION AND FUTURE WORK

In this paper, we presented a computer-aided design method that was inspired by the traditional perforated boneless lantern craft in which everyday lamps are created that contain the customized visual information of shapes and patterns. Our method effectively reduced non-creative tasks, while allowing users to engage in creative design, a manual assembly process and the interaction with a lamp.

In the future, we plan to fine-tune the parametric setting using Grasshopper to make the digital system more user-friendly for further user study and consider complex interaction conditions further (e.g., how this method can promote everyday interaction and multi-person handicraft activity).

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REFERENCES

- [1] Danny Leen, Nadya Peek, and Raf Ramakers. 2020. LamiFold: Fabricating Objects with Integrated Mechanisms Using a Laser cutter Lamination Workflow. In Proceedings of the 33rd Annual ACM Symposium on User Interface Software and Technology. Association for Computing Machinery, New York, NY, USA, 304–316. Retrieved July 14, 2021 from <https://doi.org/10.1145/3379337.3415885>
- [2] Lijuan Liu, Yang Chen, Pinhao Wang, Yizhou Liu, Caowei Zhang, Xuan Li, Cheng Yao, and Fangtian Ying. 2018. Papercut: Digital Fabrication and Design for Paper Cutting. In Extended Abstracts of the 2018 CHI Conference on Human Factors in Computing Systems (CHI EA '18). Association for Computing Machinery, New York, NY, USA, 1–6. DOI:<https://doi.org/10.1145/3170427.3188522>
- [3] Yiyue Luo, Kui Wu, Tomás Palacios, and Wojciech Matusik. 2021. KnitUI: Fabricating Interactive and Sensing Textiles with Machine Knitting. In Proceedings of the 2021 CHI Conference on Human Factors in Computing Systems. Association for Computing Machinery, New York, NY, USA, 1–12. Retrieved July 14, 2021 from <https://doi.org/10.1145/3411764.3445780>
- [4] Mako Miyatake, Koya Narumi, Yuji Sekiya, and Yoshihiro Kawahara. 2021. Flower Jelly Printer: Slit Injection Printing for Parametrically Designed Flower Jelly. In Proceedings of the 2021 CHI Conference on Human Factors in Computing Systems. Association for Computing Machinery, New York, NY, USA, 1–10. Retrieved July 14, 2021 from <https://doi.org/10.1145/3411764.3445346>
- [5] Stefanie Mueller, Bastian Kruck, and Patrick Baudisch. 2013. LaserOrigami: laser-cutting 3D objects. In Proceedings of the SIGCHI Conference on Human Factors in Computing Systems. Association for Computing Machinery, New York, NY, USA, 2585–2592. Retrieved July 14, 2021 from <https://doi.org/10.1145/2470654.2481358>
- [6] Ticha Sethapakdi, Daniel Anderson, Adrian Reginald Chua Sy, and Stefanie Mueller. 2021. Fabricaide: Fabrication-Aware Design for 2D Cutting Machines. In Proceedings of the 2021 CHI Conference on Human Factors in Computing Systems. Association for Computing Machinery, New York, NY, USA, 1–12. Retrieved July 14, 2021 from <https://doi.org/10.1145/3411764.3445345>
- [7] Haruki Takahashi and Jeeun Kim. 2019. 3D Pen + 3D Printer: Exploring the Role of Humans and Fabrication Machines in Creative Making. In Proceedings of the 2019 CHI Conference on Human Factors in Computing Systems - CHI '19, ACM Press, Glasgow, Scotland UK, 1–12. DOI:<https://doi.org/10.1145/3290605.3300525>
- [8] Ye Tao, Guanyun Wang, Caowei Zhang, Nannan Lu, Xiaolian Zhang, Cheng Yao, and Fangtian Ying. 2017. WeaveMesh: A Low-Fidelity and Low-Cost Prototyping Approach for 3D Models Created by Flexible Assembly. In Proceedings of the 2017 CHI Conference on Human Factors in Computing Systems, ACM, Denver Colorado USA, 509–518. DOI:<https://doi.org/10.1145/3025453.3025699>
- [9] Ye Tao, Nannan Lu, Caowei Zhang, Guanyun Wang, Cheng Yao, and Fangtian Ying. 2016. CompuWoven: A Computer-Aided Fabrication Approach to Hand-Woven Craft. In Proceedings of the 2016 CHI Conference Extended Abstracts on Human Factors in Computing Systems, ACM, San Jose, CA, USA, 2328–2333. DOI:<http://dx.doi.org/10.1145/2851581.2892293>